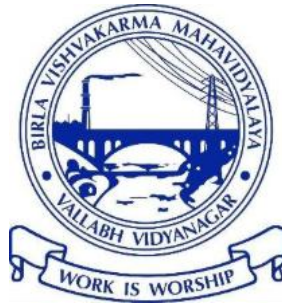


A PROJECT REPORT
ON
AI-Enhanced Decentralized Identity Verification System

Submitted by
Akshay Gojiya (22IT452)
For Partial Fulfillment of the Requirements for Bachelor of
Technology in
Information Technology

Guided by
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CERTIFICATE

This is to certify that the project work entitled **AI-Enhanced Decentralized Identity Verification System** has been successfully carried out by **Akshay Gojiya (22IT452)** for the subject **Major Project (4IT33)** during the academic year 2025-26, Semester-VIII for partial fulfilment of Bachelor of Technology in Information Technology. The work carried out during the semester is satisfactory.

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Internship Experience Letter

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To Whom It May Concern,

This is to certify that Akshay Gojiya (22IT452) of 4th Year, B.Tech. in Information Technology, Birla Vishvakarma Mahavidyalaya, Vallabh Vidyanagar, has successfully completed his project titled "AI-Enhanced Decentralized Identity Verification System" Using Blockchain and Self-Sovereign Identity under the supervision of Brijesh Patadiya at devx AI Labs.

The project focused on developing a decentralized identity system using AI and Blockchain, featuring MetaMask-based authentication, DID creation, secure document handling, AI verification, and blockchain-based credentials.

He contributed to core modules including Authentication, DID Management, Blockchain Integration, and Dashboard Development.

We appreciate his efforts and wish him success in his future endeavors. We appreciate the efforts and commitment the intern is showing in work and wish him all the best for future professional endeavors.


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Finally, I express my gratitude to the institution for giving me the platform to apply theoretical knowledge into practical implementation, thereby enhancing my technical and problem-solving skills.

Abstract

The AI-Enhanced Decentralized Identity Verification System is designed to provide a secure, privacy-preserving, and user-controlled digital identity solution. Traditional identity verification systems rely on centralized databases, making them vulnerable to data breaches and misuse of sensitive information. This project addresses these challenges by leveraging blockchain technology, artificial intelligence, and decentralized storage.

The system enables users to create a Decentralized Identifier (DID) linked to their blockchain wallet, eliminating the need for traditional password-based authentication. Identity verification is enhanced using AI techniques such as face recognition, liveness detection, and OCR-based document validation. Verified credentials are securely stored as hashes on the blockchain, ensuring immutability and trust.

Additionally, the system supports selective disclosure of identity attributes using QR codes, allowing users to share only the required information while maintaining privacy. The architecture integrates modern technologies including Next.js, Node.js, MongoDB, Ethereum blockchain, and IPFS.

Overall, the system provides a scalable, secure, and efficient approach to digital identity management, reducing dependency on centralized authorities and empowering users with full control over their identity data.

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List of Abbreviations

Abbreviation	Meaning
DID	Decentralized Identifier
SSI	Self-Sovereign Identity
VC	Verifiable Credential
ZKP	Zero-Knowledge Proof
IPFS	InterPlanetary File System
KYC	Know Your Customer
OCR	Optical Character Recognition
AI	Artificial Intelligence
ML	Machine Learning
DID Registry	Smart Contract for managing DIDs
Credential Registry	Smart Contract for storing credential hashes

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